

AWS us-east-1, 19–20 Oct 2025

The DNS race that emptied DynamoDB's endpoint

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Primary source: AWS, Summary of the Amazon DynamoDB Service Disruption in the Northern Virginia (US-EAST-1) Region, <https://aws.amazon.com/message/101925/>

Nuances: AWS re:Invent 2025, DAT453. All times PDT (UTC−7).

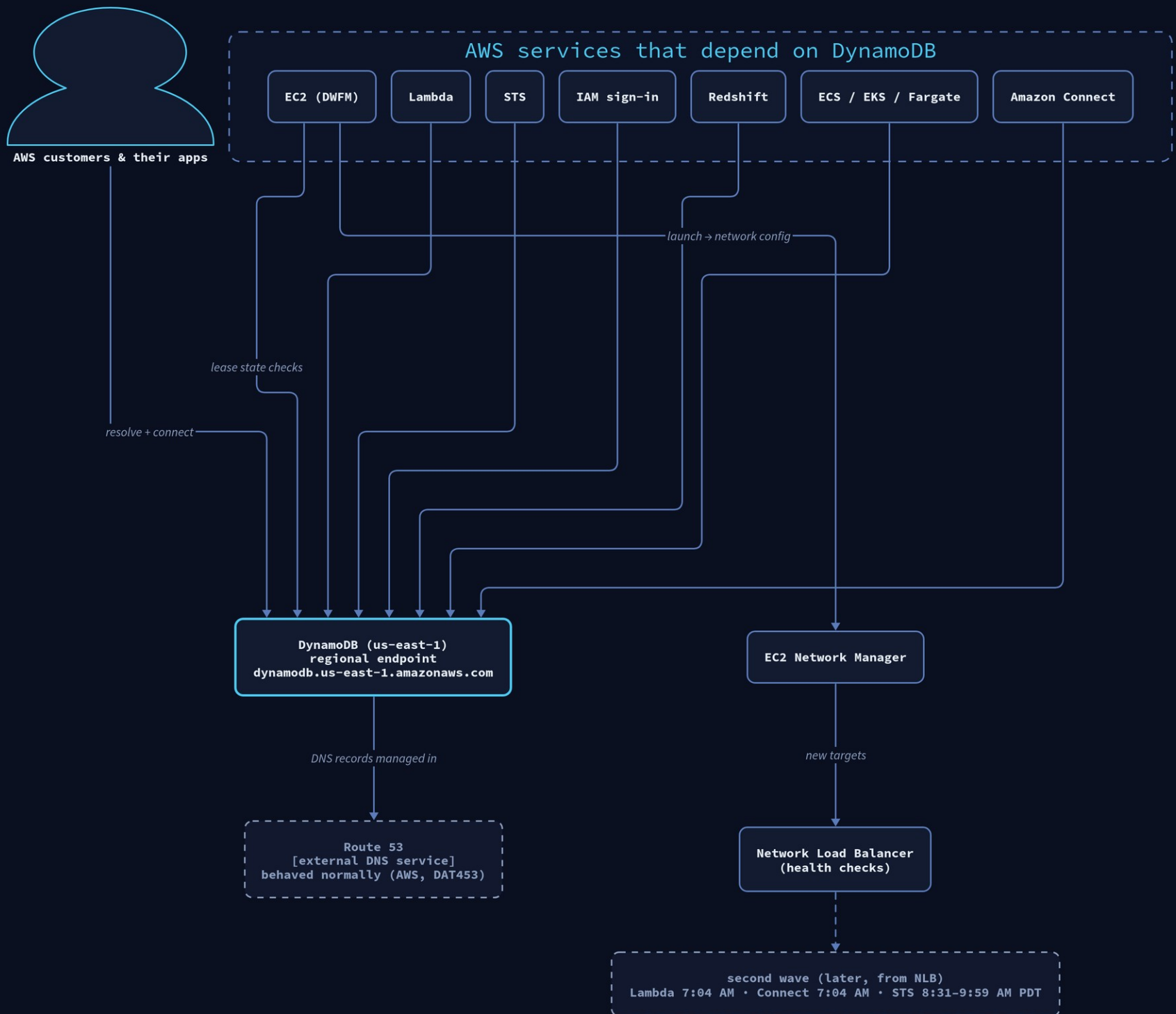
The lesson: a check that was true when it ran, a cleanup that could delete the live plan, and a recovery that became the failure mode. DNS was back by 2:25 AM; EC2 needed until 1:50 PM.

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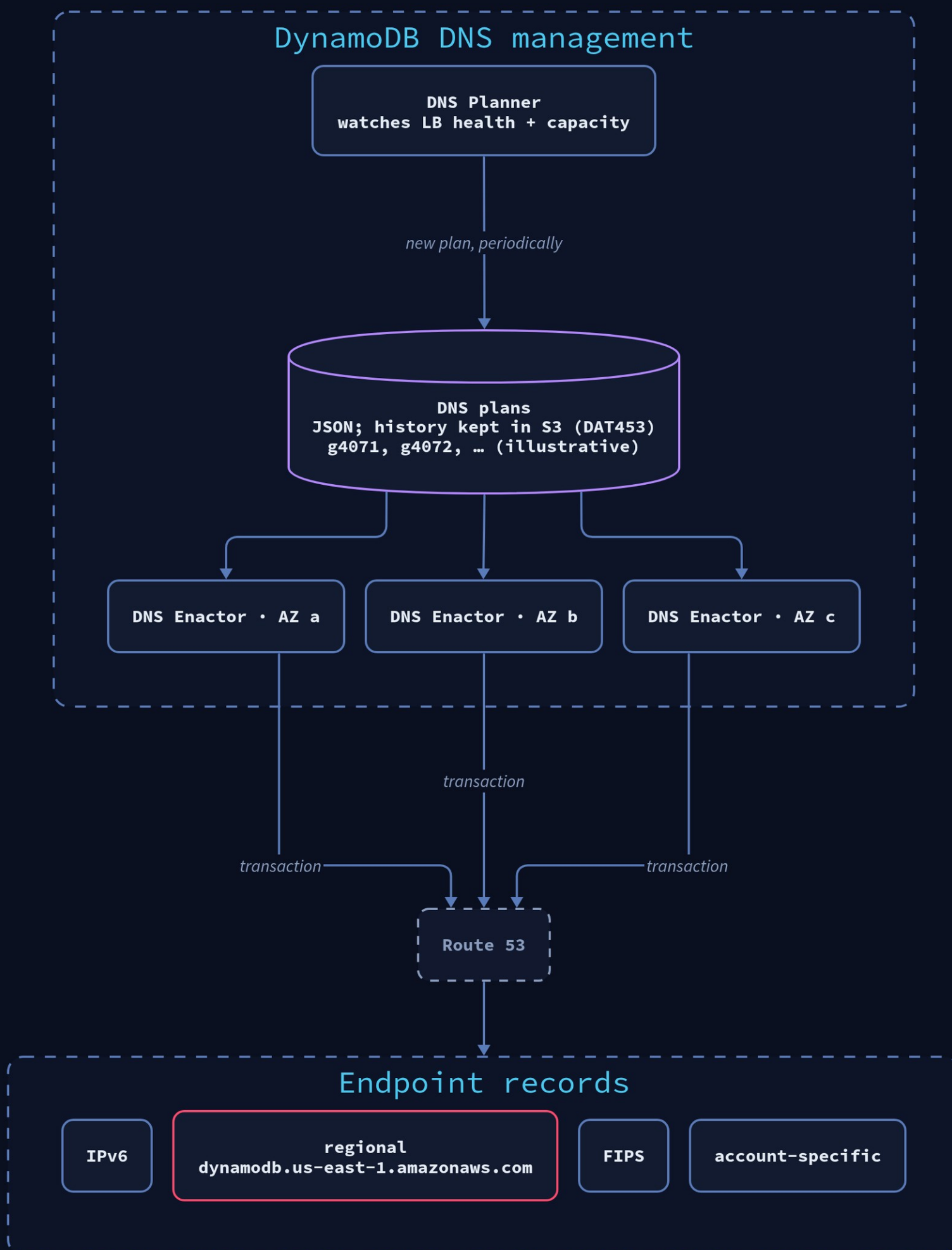
C4 level 1: the regional endpoint and who depends on it

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C4 level 2: DNS Planner, three Enactors, Route 53

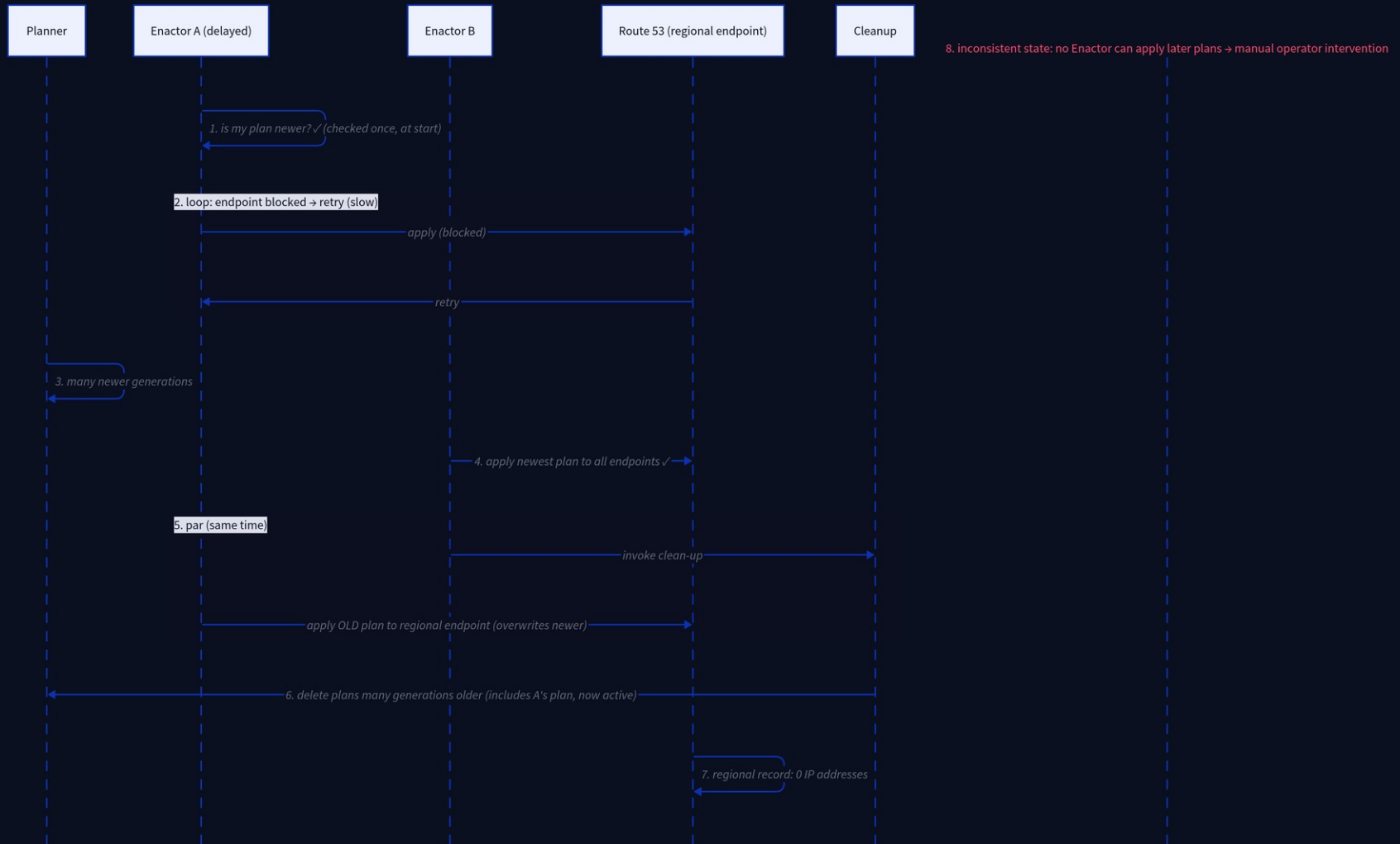
src/c4_containers_dns.d2 · svg/c4_containers_dns.svg · png/c4_containers_dns.png



one regional DNS plan, shared across endpoints (per AWS).
AWS describes the regional endpoint's record being emptied;
the other endpoints' status isn't described.

UML sequence: the race that emptied the record

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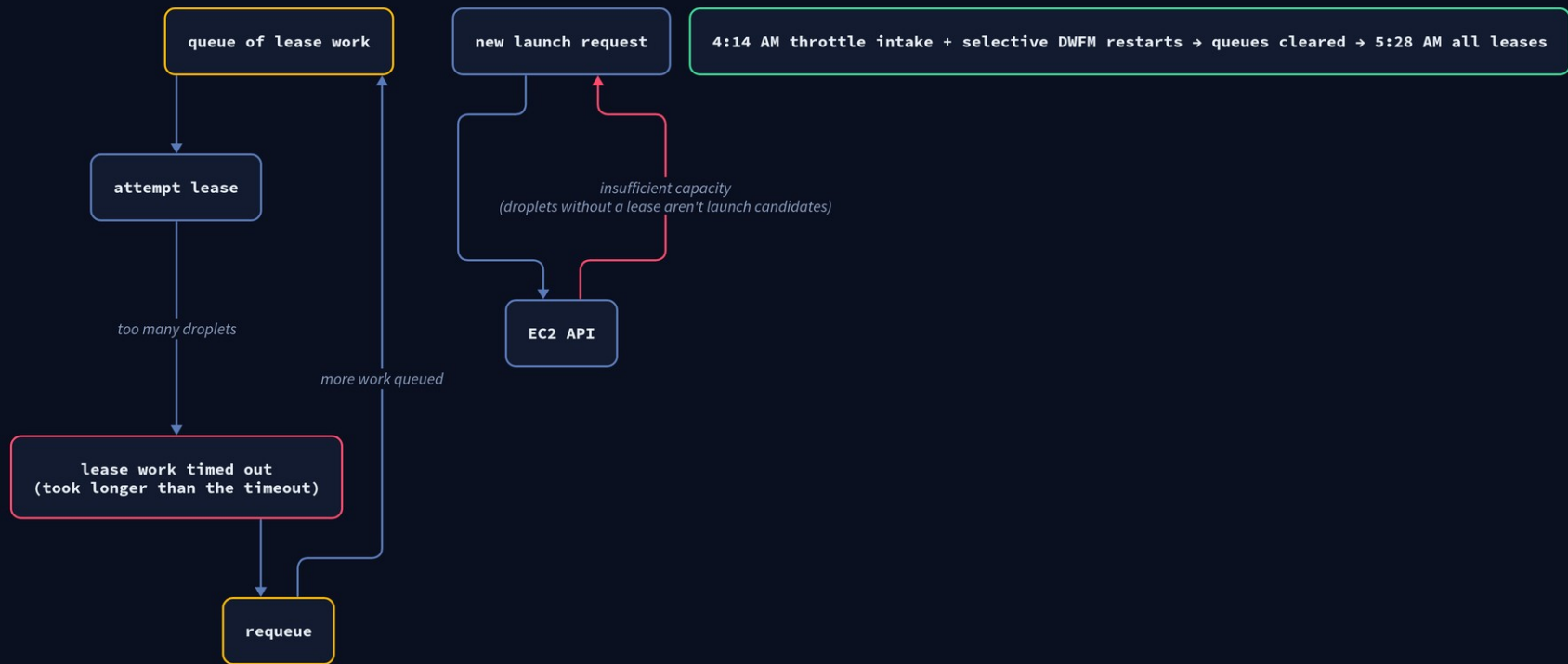
C4 level 2: DWFM, droplets, leases, Network Manager, NLB

src/c4_containers_ec2.d2 · [svg/c4_containers_ec2.svg](#) · [png/c4_containers_ec2.png](#)



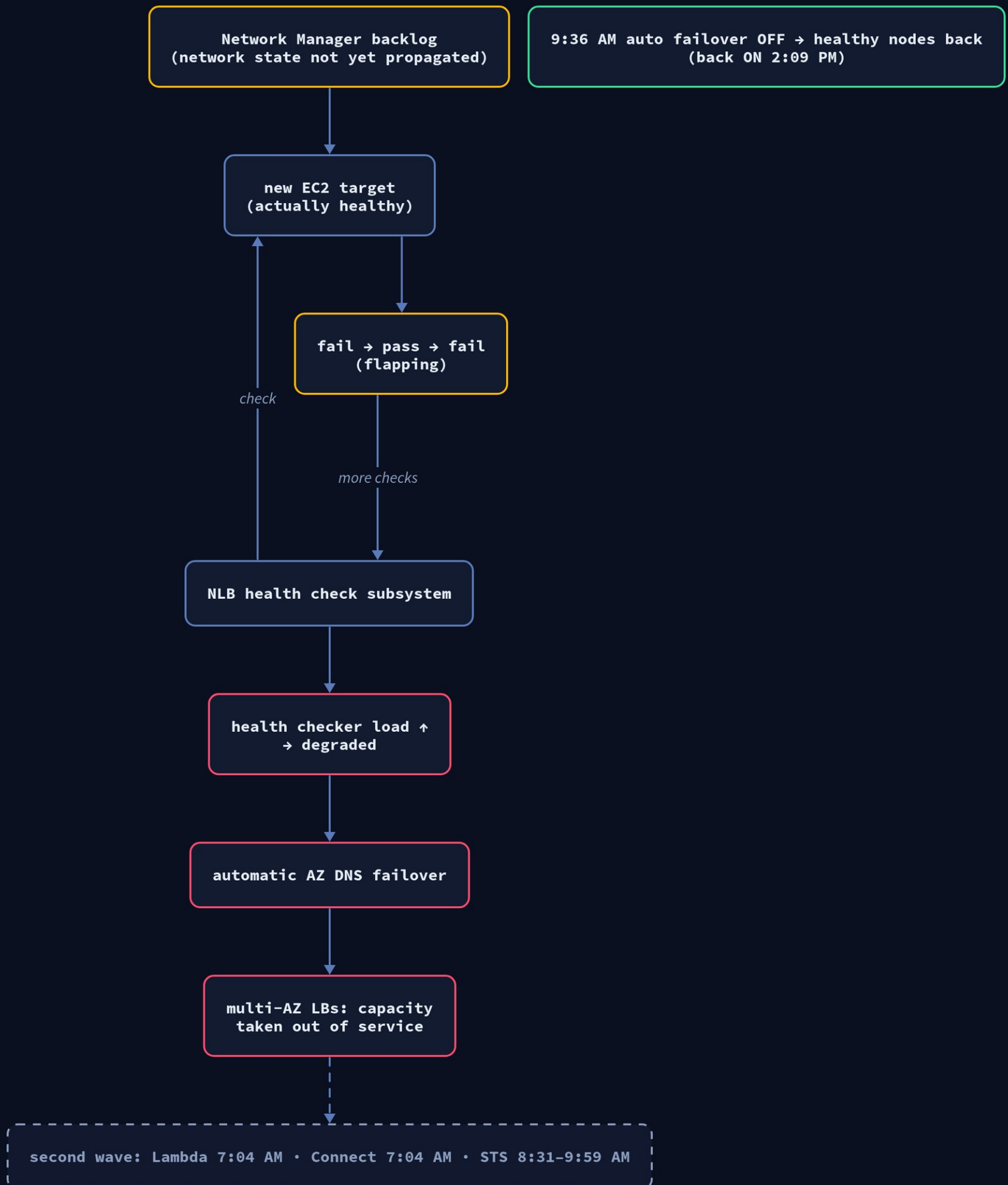
Loop: DWFM congestive collapse after 2:25 AM

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Loop: NLB health-check flapping

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Timeline: three overlapping periods of impact (PDT)

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Failure dynamics: first event, signal, shared resource, containment, trade-off, recovery

[src/failure_dynamics.d2](#) · [svg/failure_dynamics.svg](#) · [png/failure_dynamics.png](#)

PDT · per AWS	1 • DNS race (DynamoDB)	2 • DWFM congestive collapse (EC2)	3 • NLB health-check flapping
what happens first	11:48 PM delayed Enactor applies an old plan; cleanup deletes it → record has 0 IPs	11:48 PM lease checks fail; leases time out over ~2.5 h (2:24 AM)	5:30 AM new instances put in service before network state propagated
signal that detects it	DNS answers with zero addresses (no error); source identified 12:38 AM	launches answered 'insufficient capacity'	6:52 AM monitoring detects health-check flapping
shared resource that runs out	the one regional endpoint record (the active plan)	DWFM time per lease vs. the timeout (queue of expiring work)	health-checker capacity → healthy NLB capacity (multi-AZ)
what contains the spread	manual DNS repair; DNS Planner + Enactor automation disabled worldwide	4:14 AM throttle incoming work + selective DWFM restarts	9:36 AM automatic AZ failover disabled for NLB
accepted trade-off	no automated DNS updates until the fix (back on in all regions by 28 Oct)	many launches get 'request limit exceeded'	no automatic failover 9:36 AM → 2:09 PM
recovery without a second wave	2:25 AM DNS restored; clients return as cached records expire	throttles relaxed gradually from 11:23 AM; EC2 normal 1:50 PM	failover re-enabled 2:09 PM, after EC2 recovered

Monday: five recommendations (our opinion), each with what it does NOT solve

[src/monday_policy.d2](#) · [svg/monday_policy.svg](#) · [png/monday_policy.png](#)

Channel recommendations (our opinion), based on AWS's summary (aws.amazon.com/message/101925)

